

System integration

A chip is more than gates, it is blocks wired together

Starter mini CPU

- Starter preset: CPU, ALU, RegFile, and Memory on the canvas
- Cpu rf_ctrl enables the register write
- Four links are still missing
- Click an output port, then a matching input, until validation turns green

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Block-diagram integrator

Wire CPU / datapath / memory-style blocks by clicking ports. Match **direction** and **type** (addr / data / ctrl / instr). Starter: a mini CPU with ALU and register file already connected — add memory.

Starter example: ALU and RegFile are wired. Finish the mini system by connecting CPU address/data to Memory and routing rdata/instr back.

Load starter example

Challenges 0/22

Pick one Quiz: block

In a system block diagram, a block usually represents...

☐ a major functional unit (CPU, ALU, memory, bus, ...)

☐ only a single NAND gate always

☐ a SPICE transistor

☐ a Git commit

Start

Show hint

Check

Next

Stop

Idle

Block diagram starter

Workbook practice

- On paper, sketch four blocks, CPU, ALU, RegFile, and Memory, and label their ports
- Draw the five datapath wires from the starter
- Add the four CPU-to-memory links and name each signal
- Explain why addr cannot wire to a data port in this teaching model
- Optionally sketch how a bus sits between CPU and memory instead of a direct link

Pitfalls to watch

- Do not wire output to output or input to input, direction matters
- Types must match; the lab rejects addr-to-data mistakes
- One input accepts only one driver; a new wire replaces the old one
- This is a conceptual integration sketch, not a full SoC generator or place-and-route tool
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish the starter until integration passes, then load the bus preset
- On paper, list the nine required edges for the complete mini system
- When you are ready